

GenAI Ops: Iterative Prompt Refinement Engine

Duration : 90 Min.

Scenario

Develop a Python script ('refine_prompt.py') that evaluates 3 revisions of a prompt, calculates a quality metric per iteration, and enforces monotonic improvement.

Target File Location & Creation

File to Create/Update: student_workspace/refine_prompt.py **Input Resource File to Inspect:** student_workspace/prompt_revisions.txt

Task Objectives

Perform the following actions inside the student_workspace directory: - Inspect prompt_revisions.txt in student_workspace/. - Create or update student_workspace/refine_prompt.py and populate it with valid analysis results.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Open and inspect prompt_revisions.txt inside student_workspace/.
3. Create or open student_workspace/refine_prompt.py in student_workspace/ and record your answers.
4. Save your changes (Ctrl + S or Cmd + S).
5. Open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your progress by running python run.py locally in the terminal.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	refine_prompt.py is syntax-valid Python	3 Marks
TC2	Evaluates prompt specificity, role assignment, and output constraints	3 Marks
TC3	Calculates quality score per iteration	3 Marks

TC4	Enforces non-decreasing score constraint across iterations	3 Marks
TC5	Writes refinement_summary.json	2 Marks
TC6	Identifies best performing prompt iteration	2 Marks
TC7	Logs delta improvement score	2 Marks
TC8	Handles single-iteration fallback gracefully	2 Marks

Total Score: 20 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.

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